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|---|--|----------|-----|
| 8 | (a) <u>nucleus</u> has constant probability of decay
per unit time / in a given time
<i>(allow 1 mark for 'cannot predict which <u>nucleus</u> will decay next')</i> | M1
A1 | [2] |
| (b) (i) count rate / activity decreases | | B1 | [1] |
| (ii) count rate fluctuates / is not smooth | | B1 | [1] |
| (c) <i>either</i> the (decay) curves are similar / same
<i>or</i> curves indicate same half-life | | B1 | [1] |